

Topological Analysis and Approximate Identification of Leading Lines in Artworks Based on Discrete Morse Theory

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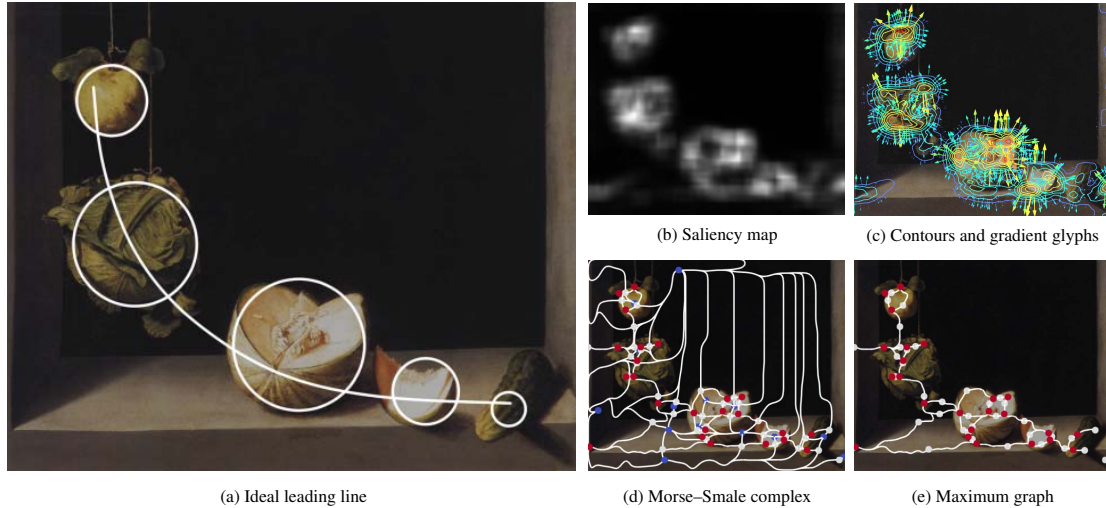


Figure 1: *Still Life With Quince, Cabbage, Melon, and Cucumber* (Juan Sánchez Cotán, c. 1602) and its compositional analysis in terms of leading lines.

ABSTRACT

Accomplished artists often incorporate leading lines into their compositions to guide the observer's attention. Although saliency maps are typically employed to locate attractive regions in still images, such scalar features do not express the trajectory of an observer's gaze. In this study, we propose a method for the visual analysis and approximate identification of leading lines based on maximum graphs, sparse subsets of Morse–Smale complexes, extracted from saliency maps. We provide empirical evidence substantiating the feasibility of our method through a comparison with actual observers' eye tracking results. Further, we investigate the limitations of our approach by employing it to analyze a variety of artworks with diverse styles.

Index Terms: Human-centered computing—Visualization—Visualization design and evaluation methods

1 INTRODUCTION

Distinguished painters and photographers often strive to draw the observer's attention to both the primary subject and the surrounding area, with the intent to encourage closer observation and appreciation of their works. One of the techniques employed to achieve this is leading lines [17], which act as pathways within a composition to guide the observer's gaze from one point to another. For instance, lines that converge are known to alter the trajectories of an observer's eyes toward a vanishing point in an image [3]. However, as exemplified in Fig. 1a, leading lines are not always delineated

explicitly; they may also be abstracted lines of sparsely arranged objects; perceived outlines, such as suggestive contours [8] that do not actually exist; or gradations and textures of brushstrokes. One challenge is that such leading lines may be difficult for observers to recognize without a background in art or photography education; yet, understanding the structure and objectifying the content of an artwork by identifying these leading lines can be valuable in art appreciation [15]. Moreover, artists' eyes can become accustomed to their own works over prolonged production periods, making it difficult even for them to identify the leading lines.

Visual attention models based on saliency are frequently employed to measure the attractive area in still images, and they were found to have been used extensively in various fields [2], including eye fixation prediction, object detection, and media quality assessment. Nevertheless, such scalar features as depicted in Fig. 1b do not capture the trajectory of gaze movements.

This study therefore presents an approach toward approximating the identification of leading lines that is accessible to novice artists and individuals interested in developing their art appreciation skills. The approach involves abstracting the gradient field of a saliency map. However, considering the visualization of the gradient field using only arrow plots can appear cluttered, as seen in Fig. 1c, we propose a method that utilizes a Morse–Smale (MS) complex (Fig. 1d) derived from a saliency map. Particularly, we focus on its sparse subset, known as the maximum graph [6], which exclusively links maxima and saddle points, as illustrated in Fig. 1e. In addition, we confirm the feasibility of this method by comparing the approximated leading lines with the eye tracking results of average observers.

2 RELATED WORKS

In this section, three prior studies are introduced, and we elucidate the direction of our research.

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Mochizuki et al. [18] introduce the concept of leading lines as a structural element that converges toward a particular point within an image. They presume the saliency of a region is non-linearly enhanced by the spatial consistency of the leading lines and the low-level center-surround differences in visual stimuli, such as color and luminance. To account for the attractiveness of leading lines, they propose a novel saliency model that applies an amplified Gaussian kernel, which reflects the synergistic effect, to the detected vanishing point. Meanwhile, Takahashi et al. [21] introduce a method for estimating the eye fixation location on an input image by identifying the intersection points of lines extended from detected edges. Conversely, Xia et al. [28] propose an approach that integrates both low-level image features, such as color and texture, and high-level image features that are semantically meaningful. They first extract graphlets to represent local features, and they then project them onto perceptual space and further decompose them into feature vectors by employing a locality-preserved, non-negative matrix factorization algorithm. Assuming humans' eye movements are driven by both visual and semantic attractiveness, they sequentially link the graphlets to infer human gaze-shifting paths, starting from the most salient regions.

In the studies conducted by Mochizuki et al. [18] and Takahashi et al. [21], only prominent lines that converge at a certain point are considered leading lines. Conversely, Xia et al. [28] investigate both low- and high-level image features to estimate gaze paths, but they are simply a sequence of segmented regions based on saliency and are only capable of estimating a single path that does not branch. Because the objective of these studies is not to identify the leading line itself, the detected gaze trajectories are limited to simple lines. In contrast, our work, based on discrete Morse theory, aims to visually analyze a more complicated leading line that can account for the sequence of spaced objects and other relevant factors.

3 APPROACH

We hypothesize that the distribution of visually appealing regions influences the direction of the human gaze. Our approach leverages discrete Morse theory to abstract the gradient field of the saliency map computed from a still image input, and it involves performing a visual analysis of the leading lines.

3.1 Saliency map

When creating an artwork, artists often emphasize important parts of the piece by manipulating the contrast or precision of the drawing, as well as by strategically placing some manipulated details to suggest a leading line. Our approach employs a saliency map, which is an image feature that accounts for human visual characteristics, to identify highly attractive regions within an input image that serve as the source of such leading lines, as depicted in Fig. 1a. To generate the saliency map in Fig. 1b, we employed a fast processing model, specifically proposed by Hou and Zhang [14].

3.2 Contour and gradient

The direction of the human gaze is influenced by the attractiveness of the visual features in view, and it is typically drawn from larger to smaller objects and from brighter to darker regions in an image. To enhance the legibility of the saliency map, we first extracted its contour lines and overlaid them onto the map, after which we computed the gradient field of the saliency map, allowing the visualization of the gaze flow patterns induced by the distribution of attractive regions using arrow plots. Figure 1c shows the contours and arrow plots of the gradient field obtained from the saliency map in Fig. 1b. Although this visualization enabled us to depict the local features of leading lines, the dense arrangement of many glyphs, which tend to overlap each other, led to visual clutter and was unsuitable to represent comprehensive leading lines.

3.3 Morse–Smale complex and maximum graph

We contemplated that the further abstraction of leading lines may be achieved by interconnecting highly salient points via lines. As an alternative to the arrow glyph representation, we decided to adopt an abstract representation by the MS complex, which is a form of differential topological data analysis that partitions a scalar field into uniformly gradient regions by interconnecting all critical points as a graph [10]. The saliency map, derived from an input image, is regarded as a 2-manifold with boundary, and the MS function over the manifold is obtained [20]. The MS complex is known to benefit the shape analysis of scalar fields on manifolds through segmentation [7]. As mentioned above, leading lines do not always form explicit lines. Moreover, our eye movements alternate along a path repeatedly while scanning an image, and the leading line can be identified as a central path that represents the bundle of gaze trajectories. Therefore, we believe the skeletal representation of segmented regions provided by 1-cells of MS complexes closely resembles the structure of leading lines.

A MS complex is obtained as a collection of MS cells, including 0-cells, representing vertices corresponding to critical points; 1-cells, representing arcs tracing half of a stable or unstable 1-manifold; and 2-cells, representing regions with uniform gradients, where stable and unstable 2-manifolds intersect. In our visual analysis method, we utilize the skeletal paths represented by these 1-cells, also known as extremum graphs [6]. When considering a meshed saliency map as input, derivation of the extremum graphs follows the paths from the saddle point along the edges of the mesh, ascending and descending the steepest local gradient at all regular points. Each path is followed by until a critical point is reached or until it intersects with the path that has already been traced. The graph that connects all critical points, using the collection of ascending and descending paths from the saddle point, is illustrated in Fig. 1d.

However, to visualize the leading lines that connect sparsely arranged salient areas, it is not necessary to include low-saliency points in the graph. Consequently, we have opted to analyze such sparse leading lines by extracting the maximum graph [6], a subgraph of the extremum graph. This subgraph, as depicted in Fig. 1e, will only connect the maxima (plotted in red) and saddle points (plotted in white) of the saliency map.

3.4 Topological simplification

Detecting all critical points in the saliency map, including minor maxima, may complicate the analysis, making it necessary to control the level of topological detail by removing minor critical points. To do so, we set a threshold on the persistence [11] of critical point pairs [12], and we extract a simplified scalar field by selectively removing minor critical points [25] [16], as shown in Fig. 2.

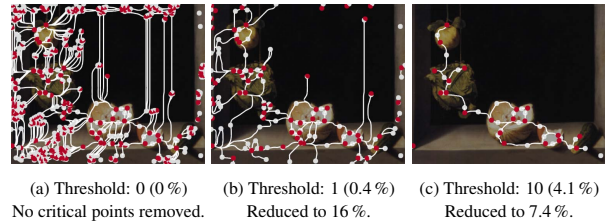
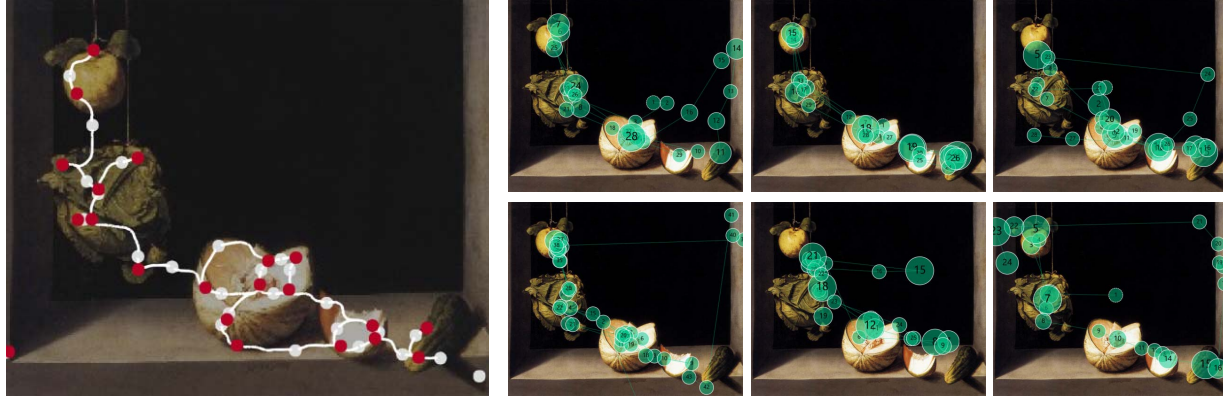


Figure 2: Results obtained through thresholding. The first percentage indicates the threshold value relative to the highest persistence value, while the second the resulting number of critical points.

Persistent homology is used to obtain geometric information of input data by pairing the creation and destruction of topological structures. Consisting of 2-manifolds, the topological persistence of critical points is determined by the lifetime of its structure in a



(a) Thresholded maximum graph

(b) Gaze trajectories of six average observers

Figure 3: A comparison of the outcomes generated by our approach when applied to the painting in Fig. 1, and the eye tracking data collected from six graduate/undergraduate students majoring in computer science. The size of each plot indicates the duration of fixation at a specific point, while the number on the plot signifies the sequence of the observed gaze trajectories. Supplementary materials that provide high-resolution versions of these results for diverse artworks are available.

scalar field. First, every maximum and the preceding saddles are paired. As the isovalue continuously decreases, new components appear at the maximum and begin to connect at saddles, resulting in changes to the topological structure. The saddle is paired with the lower maximum, and the highest maximum continues to represent the integrated component. The isovalue at which a component is generated is referred to as the birth time, while the isovalue at which it connects and disappears is called the death time; together, they are known as a birth–death pair.

The two graphs on the right-hand side of Fig. 4 illustrate the persistence of the critical points. The set of birth–death pairs is plotted in the lower graph, called the persistence diagram [9]. The horizontal axis indicates the birth time, while the vertical axis indicates the death time, whereas length of each bar represents the persistence of each birth–death pair. The upper graph, showing the persistence curve, describes the number of critical point pairs at each persistence level. By setting a threshold on persistence and selectively removing the minor maximum based on these graphs, all vertices of a low persistence feature can be flattened to the closest saddle [16] for pre-simplification of the saliency map.

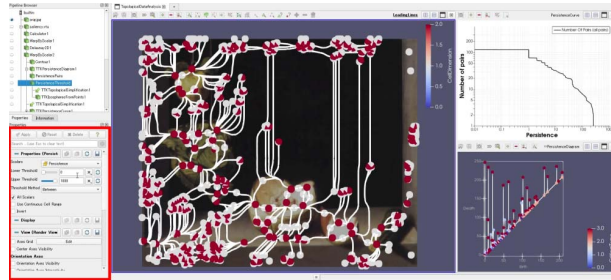


Figure 4: A screenshot of the graphical user interface used for the visual analysis of leading lines. The threshold value is adjusted in the panel enclosed in red on the left side of the screen.

4 EXPERIMENTS

We developed a visual analysis system for leading lines by utilizing Python 3.9.15 in combination with OpenCV [4], the Visualization Toolkit (VTK) [19], and the Topology ToolKit (TTK) [24] libraries.

Figure 4 is a screenshot of the interface that was implemented using Paraview 5.10.1 [1], running on a 64-bit Windows 11 with an Intel Core i7-11700KF 3.60 GHz processor and 128 GB of RAM.

4.1 Approximating leading lines

Figure 3a depicts the output results of our approach when the painting in Fig. 1 is fed as the input. Concerning the current framework, the saliency maps are downsized to 256×256 px or less, regardless of the original resolution of the input image, as the number of detected critical points is restricted to a maximum of 10,000 pairs in our implementation. After the saliency map is computed, the intensity and contrast of the background reference image are reduced from the original input to enhance the visibility of the maximum graph. Deriving the MS function requires less than 10 seconds, and the subsequent thresholding process can be carried out interactively on Paraview, as demonstrated in Fig. 4.

4.2 Eye tracking results

We used the Tobii Pro Spectrum eye tracking device with the Tobii Pro Labs software [26] to monitor the eye movements of six actual observers, who are all students majoring in computer science. The images were presented for a duration of 10 seconds each on a full screen with a 23.8-in (53.8×33.2 cm) display, with a resolution of $1,920 \times 1,080$ px, at a distance of 60 cm from the observers.

The graph that was identified from the painting, with relatively simple compositions, depicted in Fig. 3a, closely approximated the ideal leading lines illustrated in Fig. 1a. Moreover, the resulting graph exhibited characteristics that were qualitatively similar to the connections of the actual fixation points demonstrated in Fig. 3b.

4.3 Examples on various styles of artworks

Figure 5 illustrates the leading lines approximated in a range of artworks by applying persistency thresholding to each example. For the representational paintings featuring multiple characters (Fig. 5b), our method proved capable of representing complicated leading lines that intersect, and the resulting graphs resembled the eye tracking results to some extent. However, the tracking results revealed that the observer’s attention was predominantly focused on the facial regions of the characters, whereas the proposed method often exhibited a deviation, and the resulting graph did not encompass the position of the face, even for the primary character of the painting, because the present approach does not account fully for the cognitive psychological context. In addition, the approach

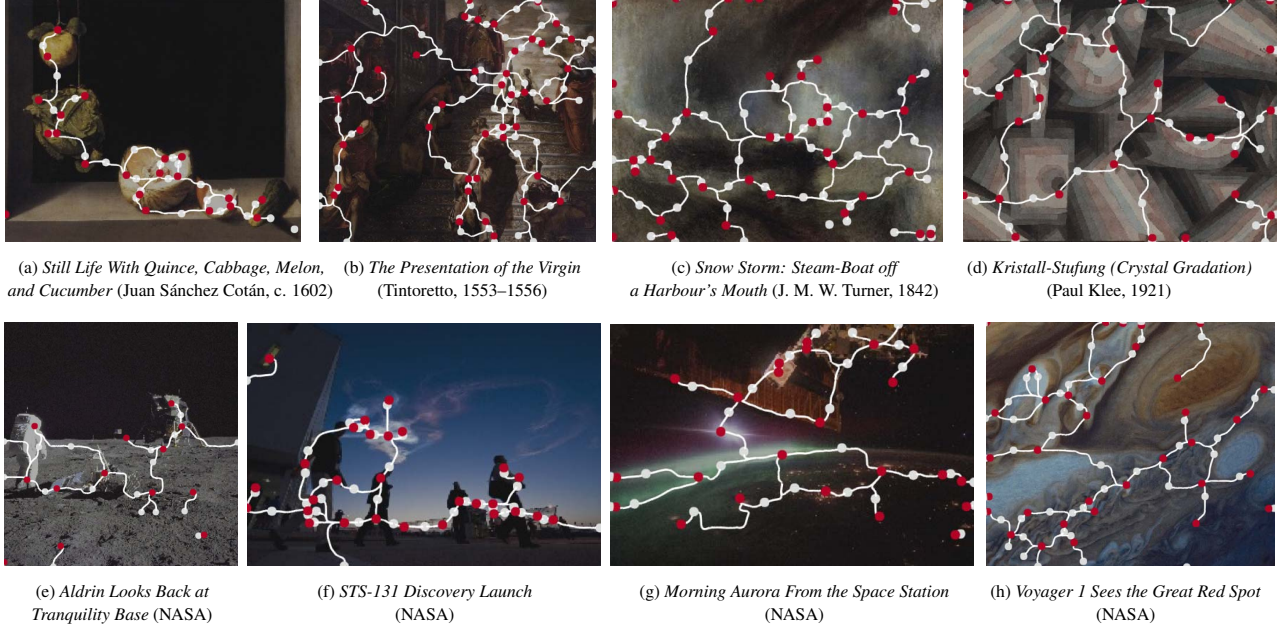


Figure 5: The leading lines approximated in various paintings and photographs using our method. The threshold values and high-resolution versions of these outputs, along with the corresponding eye tracking results for each image, are available as the supplementary materials.

was unable to approximate the leading lines in works like Fig. 5c, where the brush strokes are considered to represent leading lines, or Fig. 5d, which combine geometric patterns. Nevertheless, a comparison of the tracking results indicated that the proposed method can approximate the leading lines, even in a wide range of photographic scenes, as presented in Fig. 5e through Fig. 5h.

5 DISCUSSIONS

Some limitations of the proposed method were identified, necessitating further investigation.

5.1 Experimental evaluation

We demonstrated the efficacy of our approach by monitoring the gazes of average observers and contrasting their gaze trajectories with the leading lines approximated using our method. However, for a more quantitative comparison, it is necessary to investigate the output results through a third-party evaluation. To delve further into this issue, we aspire to conduct evaluation experiments with students from art universities and with visual psychology experts.

5.2 Good initial guess of the threshold value

Although the objective of our study is to offer a tool for people without a background in art education to identify leading lines in artworks, our approach necessitates the user to set a threshold on the persistence value. This is due to the difficulty in determining the optimal threshold value for identifying leading lines. However, it may be feasible to provide a good initial guess for thresholding by applying the elbow method [23], a widely known technique for determining the number of clusters in a dataset, to the persistence curve.

5.3 Identifying the eye trajectory orientation

Currently, the orientation of a gaze shift along approximated leading lines remains unexamined, but it may be possible to estimate it based on the disparity in saliency between each maxima or on the corresponding area of the ridge cycle [22] of the saliency map.

Furthermore, there may be potential to determine the orientation by utilizing vector field topology [13], and in the future, we intend to compare the outcomes with those based on the maximum graph.

5.4 Multivariate analysis

As the proposed method is based on a saliency model that employs local low-level image features, it can approximate the leading lines that align with the saccade direction [27]. However, it is important to note that content-based leading lines, such as those conveyed through people's gestures and facial expressions, which may deviate from the saccade direction, cannot be captured within the current framework. In addition, some leading lines based on low-level image features, such as brush strokes, also remain elusive. To overcome these issues, the utilization of multivariate computational topology techniques, such as contour nets [5], is required, as they enable the analysis of relationships among various properties, including other low-level image features and semantic information.

6 CONCLUSION

In this paper, we demonstrated the validity of maximum graphs derived from saliency maps for the visual analysis and approximate identification of leading lines in artistic compositions. Furthermore, we conducted a comparative study on the approximated leading lines using a variety of paintings and photographic works along with eye tracking results obtained from real observers. To the best of our knowledge, this is the first report on leading line approximation utilizing discrete Morse theory.

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